

Introduction to Psychology

Lecture Notes — October 2023

Department of Psychology | Undergraduate Course

These lecture notes accompany the spoken lecture recordings for the Introduction to Psychology course. Each section below corresponds to a topic covered during the lecture session. Students are encouraged to supplement these notes with their own observations and the lecturer's verbal explanations.

1. What is Psychology?

Psychology is the scientific study of the mind and behaviour. It encompasses a broad range of topics including how people think, feel, act, and interact with one another. As a discipline, psychology sits at the intersection of natural science, social science, and the humanities.

The word 'psychology' derives from the Greek words 'psyche' (meaning soul or mind) and 'logos' (meaning study or reason). While early conceptions of psychology were rooted in philosophy, the field became a formal scientific discipline in the late 19th century.

1.1 Core Goals of Psychology

Psychologists typically pursue four main goals in their work:

- **Description:** Observing and describing behaviour as accurately as possible.
- **Explanation:** Understanding why a behaviour occurs and identifying its causes.
- **Prediction:** Anticipating how people will behave in future situations.
- **Control:** Applying psychological knowledge to change or improve behaviour and mental processes.

1.2 Psychology vs Common Sense

Many people believe that psychology simply confirms what we already know through common sense. However, psychological research frequently challenges our intuitions. For example, research shows that punishment is often less effective than reward in shaping behaviour — a finding that contradicts many everyday assumptions about discipline.

2. A Brief History of Psychology

The history of psychology reflects changing views about the nature of the mind and the appropriate methods for studying it. Understanding this history helps us appreciate why the field looks the way it does today.

2.1 Wilhelm Wundt and Structuralism

In 1879, Wilhelm Wundt established the first formal psychological laboratory in Leipzig, Germany. This event is often cited as the birth of psychology as an independent scientific discipline. Wundt's approach, known as structuralism, sought to understand the structure of the mind by breaking conscious experience into its most basic elements.

Wundt used a method called introspection — training participants to carefully observe and report their own mental processes. While introspection had limitations, it represented the first systematic attempt to study the mind scientifically.

2.2 William James and Functionalism

American psychologist William James proposed an alternative approach called functionalism. Rather than focusing on the structure of the mind, James was interested in how mental processes function — that is, how they help organisms adapt to their environment. James was heavily influenced by Charles Darwin's theory of evolution.

James authored the influential text 'Principles of Psychology' (1890), which remained a standard reference for decades. His emphasis on the practical purpose of mental processes laid the groundwork for applied psychology.

2.3 Sigmund Freud and Psychoanalysis

Sigmund Freud, a Viennese physician, developed psychoanalysis — a theory and therapeutic approach centred on the role of unconscious mental processes. Freud argued that much of human behaviour is driven by unconscious desires, conflicts, and memories, particularly those rooted in childhood experiences.

While many of Freud's specific claims have not held up to scientific scrutiny, his emphasis on the unconscious mind, defence mechanisms, and the importance of early experiences has had a lasting influence on psychology and popular culture.

3. Major Perspectives in Psychology

Modern psychology is not a single unified field but a collection of perspectives, each offering a different lens through which to understand human behaviour and mental processes.

3.1 The Biological Perspective

The biological perspective examines how genetics, brain structure, neurotransmitters, hormones, and the nervous system influence behaviour and mental processes. Advances in brain imaging technology such as MRI and fMRI have made this one of the fastest-growing areas of psychology.

Key questions include: How do genes influence personality? What brain regions are involved in emotion? How do chemical imbalances contribute to mental disorders?

3.2 The Psychodynamic Perspective

Rooted in Freud's work, the psychodynamic perspective emphasises the role of unconscious processes and early childhood experiences in shaping personality and behaviour. Modern psychodynamic theorists have revised many of Freud's original ideas while retaining the emphasis on internal mental conflict and the importance of relationships.

3.3 The Behavioural Perspective

Behaviourism, pioneered by John B. Watson and later B.F. Skinner, argued that psychology should focus exclusively on observable behaviour rather than internal mental states. Behaviourists emphasised the role of environmental stimuli and reinforcement in shaping behaviour.

Classical conditioning (Pavlov's experiments with dogs) and operant conditioning (Skinner's work with reinforcement and punishment) are foundational concepts from this perspective.

3.4 The Cognitive Perspective

The cognitive perspective focuses on internal mental processes such as perception, memory, thinking, language, and problem-solving. Cognitive psychologists study how people process, store, and retrieve information. This perspective emerged as a reaction to behaviourism's exclusion of mental processes.

3.5 The Humanistic Perspective

Humanistic psychology, associated with Abraham Maslow and Carl Rogers, emphasises human potential, free will, and the importance of subjective experience. Maslow's hierarchy of needs and Rogers' concept of unconditional positive regard are key contributions from this perspective.

The humanistic perspective offered an optimistic counterpoint to psychoanalysis and behaviourism, focusing on personal growth, self-actualisation, and the inherent goodness of people.

4. Research Methods in Psychology

Psychology is a science, and like all sciences it depends on systematic observation and evidence. Understanding research methods is essential for evaluating psychological claims critically.

4.1 The Scientific Method

The scientific method involves a cycle of observation, hypothesis formation, testing, and revision. In psychology, this typically means: identifying a research question, reviewing existing literature, formulating a testable hypothesis, collecting data, analysing results, and drawing conclusions.

4.2 Experimental Research

Experiments are the gold standard of psychological research because they allow researchers to establish cause and effect relationships. In a controlled experiment, the researcher manipulates an independent variable and measures its effect on a dependent variable, while controlling for extraneous variables.

Random assignment of participants to conditions is critical — it ensures that any differences between groups at the start of the study are due to chance rather than systematic bias.

4.3 Correlational Research

Correlational research examines the relationship between two variables without manipulating either. A correlation coefficient (ranging from -1 to +1) indicates the strength and direction of the relationship. However, correlation does not imply causation — a classic principle in psychological research that students must always keep in mind.

4.4 Ethical Considerations

Psychological research involving human participants must adhere to strict ethical guidelines. These include: obtaining informed consent, protecting participant confidentiality, minimising harm, allowing participants to withdraw at any time, and debriefing participants after any deception used in the study.

Institutional Review Boards (IRBs) or Ethics Committees review research proposals to ensure they meet these standards before data collection begins.

5. Biological Bases of Behaviour

Understanding behaviour requires understanding the biological systems that underlie it, particularly the nervous system and the brain.

5.1 Neurons and Neural Communication

The nervous system is made up of billions of neurons — specialised cells that transmit information throughout the body. Each neuron consists of a cell body, dendrites (which receive signals), and an axon (which transmits signals to other neurons).

Neurons communicate across synapses through chemical messengers called neurotransmitters. Different neurotransmitters are associated with different functions: for example, dopamine is linked to reward and motivation, serotonin to mood regulation, and acetylcholine to muscle movement and memory.

5.2 The Brain

The human brain is the most complex organ in the known universe. It is divided into several key structures, each associated with different functions:

- **The brainstem:** Controls basic life functions such as breathing, heart rate, and sleep.
- **The cerebellum:** Coordinates movement and balance.
- **The limbic system:** Involved in emotion, memory, and motivation. Includes the hippocampus and amygdala.
- **The cerebral cortex:** The outer layer of the brain, responsible for higher-order functions such as thinking, language, and perception.

5.3 The Endocrine System

Alongside the nervous system, the endocrine system regulates behaviour through hormones — chemical messengers released into the bloodstream. Key glands include the adrenal glands (which release adrenaline during stress) and the pituitary gland (often called the 'master gland' because it regulates other glands).

6. Sensation and Perception

How do we make sense of the world around us? The study of sensation and perception addresses this fundamental question.

6.1 The Difference Between Sensation and Perception

Sensation refers to the raw data collected by our sensory organs — the physical stimulation of our eyes, ears, skin, and so on. Perception refers to the process by which we organise, interpret, and give meaning to that sensory data.

For example, when you hear a piece of music, sensation involves the detection of sound waves by your ears. Perception involves recognising it as a melody, identifying the instrument, and perhaps feeling an emotional response.

6.2 Perceptual Organisation

Gestalt psychologists identified several principles by which the brain organises sensory information into meaningful patterns. These include figure-ground organisation (distinguishing an object from its background), proximity (grouping nearby elements together), and similarity (grouping similar elements together).

6.3 Top-Down vs Bottom-Up Processing

Bottom-up processing begins with raw sensory data and builds up to a complete perception. Top-down processing uses prior knowledge, expectations, and context to interpret sensory input. In practice, both processes work simultaneously and interact with one another.

Summary and Key Takeaways

This lecture introduced the foundational concepts of psychology as a scientific discipline. The key points covered were:

- Psychology is the scientific study of the mind and behaviour, with four core goals: describe, explain, predict, and control.
- The field has evolved through several major historical phases, from structuralism and functionalism to psychoanalysis and beyond.
- Modern psychology includes multiple perspectives — biological, psychodynamic, behavioural, cognitive, and humanistic — each offering complementary insights.
- Research methods including experiments and correlational studies form the empirical backbone of the field.
- Biological systems — particularly the brain and nervous system — are fundamental to understanding all behaviour.
- Sensation and perception are distinct but interrelated processes through which we experience the world.

Next lecture: We will explore states of consciousness, including sleep, dreams, and altered states. Please read Chapter 3 of the course textbook before the next session.